

Crabs on camera: code documentation

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2026-08-05

Background

Project and repository aims

The basis of this project is rooted in the work done in the masters thesis by Brøten (2026). The aim of the project was to improve monitoring of the data-limited Brown crab (*Cancer pagurus*) stock in Norway using machine learning and underwater video surveys.

The aim of this repository is not to push the boundaries of machine learning or showcase advanced coding, rather to make these methods more approachable for marine scientists working in fisheries applications.

The brown crab supports a valuable small-scale inshore fishery of ecological, economic and social importance to coastal communities. Despite this, data limitations persist and constrain stock assessments. Traditional methods for collecting fisheries independent data are manual, and prone to bias (Hiddink *et al.* 2024). The Institute of Marine Research (IMR) already runs coastal monitoring programs and surveys that collect underwater video footage across many sites along the coast. This footage is a large, currently underutilized resource for stock monitoring, but a bottleneck persists due to the manual annotation process (Durden *et al.* 2016). This project explores whether a machine learning model can automate that review step, using video data collected through IMR's existing coastal monitoring programs and surveys to train a crab detector.

About the repository

This repository uses the same code, functions, and packages developed for the masters thesis, though some functions have been simplified here to make the pipeline easier to reproduce.

To try it yourself, clone this repository — either on your own computer for the small example dataset, or on a server with GPU access if you plan to train on a larger dataset.

For more in depth information about the sampling methods and areas, annotation choices and a discussion of methods can be found in Brøten (2026). AI tools (large language models) were used throughout this repository to help generate code, fix syntax, and clean up errors.

Before running the scripts

Prerequisites

Python must be installed. `raw_to_yolo.ipynb` checks for missing packages and installs them automatically (see Chunk 0).

A server with GPU access is recommended if you plan to train on a large dataset, the small example dataset shipped with this repo can be run locally without one.

If you want to train a detector on your own data, you'll need an annotated dataset structured the same way as the example dataset: an `annotations` folder containing the CVAT/COCO export (`instances_default.json`), and an `images/default` folder containing the matching image files.

Data

The repo only ships a small example/test dataset (`data/annotations/test_annotations`) so the scripts can be run and verified without downloading everything.

The dataset used here for testing is from the Institute of Marine Research inshore gill- and fyke-net survey conducted in 2025. The material is from five Baited Remote Underwater Video (BRUV) stations, and covers 5 stations over 541 images.

The images are labelled with multiple identifiers to make analysis and stratified sampling later simpler. Ex. `NO_2025_3007_GarnRuse_Lovund_LO3_RA1_L9_frame_9`, is collected in Norway, in 2025 on the 30.07 on the Garn and Ruse Survey, the location name is Lovund, and its station number is LO3, and the BRUV-rig used is L9, and this is video number 9 from the left camera (L9), and is frame number 9.

From raw annotations to YOLO-ready data

Chunk 0-2

The first chunks download and load the necessary packages

References

- Brøten G. Crabs on camera: Improving the monitoring of the brown crab (*Cancer pagurus*) with machine learning and Baited Remote Underwater Video (BRUV). Master's thesis, University of Bergen, 2026. <https://hdl.handle.net/11250/5537346>.
- Durden J, Bett B, Schoening T *et al.* Comparison of image annotation data generated by multiple investigators for benthic ecology. *Marine Ecology Progress Series* 2016;**552**:61–70. <https://doi.org/10.3354/meps11775>.
- Hiddink JG, Coleman MT, Brouwer S *et al.* Estimating the abundance of benthic invertebrates from trap-catch data. *ICES Journal of Marine Science* 2024;**81**(1):86–96. <https://doi.org/10.1093/icesjms/fsad178>.
- Sekachev B, Manovich N, Zhiltsov M *et al.* *Opencv/Cvat: V1.1.0*. 2020. <https://zenodo.org/records/4009388> (7 Aug. 2026, date last accessed).